SDG 15: Life on Land

Inclusive Ecosystem Protection through Biodiversity and Land Conservation Communication

Introduction

Sustainable Development Goal 15 (SDG 15) aims to protect, restore, and promote the sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt land degradation, and protect biodiversity. In Sri Lanka, forests, wetlands, homegardens, protected areas, and wildlife habitats are essential for livelihoods, flood regulation, food security, and ecological resilience.

Sri Lanka is recognized as one of the world's biodiversity hotspots and supports more than 3,200 endemic species across a relatively small landmass. However, endemic species and habitats face pressure from habitat fragmentation, human-wildlife conflict, development, tourism pressure, land-use change, and uneven conservation attention.

Wetland degradation is also a major concern. Research from Kolonnawa shows how rapid urbanization, settlement expansion, land reclamation, and poor drainage have reduced wetland functions that once helped manage floodwaters. Community members may understand the value of wetlands, but top-down restoration plans can create fear of displacement and livelihood disruption.

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Modelled projections suggest that language-accessible ecosystem communication could ***improve community understanding of conservation and restoration guidance by 40%, increase wildlife-conflict reporting and support-service inquiry completion by 25%, reduce confusion around protected-area rules and wetland restoration plans, and strengthen participation in biodiversity awareness programs.*** This supports SDG Targets 15.1, 15.2, 15.3, 15.5, 15.7, and 15.9.

Use Context: Language Barriers in Biodiversity Protection, Wetland Restoration and Human-Wildlife Coexistence

Achieving SDG 15 in Sri Lanka requires more than declaring protected areas or designing restoration projects. Communities must be able to understand forest-use rules, wetland protection guidance, wildlife safety messages, biodiversity education, compensation pathways, ecotourism instructions, land restoration activities, and reporting mechanisms. This is especially important for rural farming communities, informal urban settlements, tourism-adjacent communities, and local groups living close to sensitive ecosystems.

Human-elephant conflict illustrates the link between environmental degradation and rural livelihoods. The source on Lunugamwehera shows that agricultural damage, property damage, socioeconomic position, psychological stress, health measures, and habitat degradation can shape household economic losses. It also notes that adaptive measures and policy implementation may be limited in effectiveness, showing the need for clearer, community-level mitigation support.

Biodiversity conservation also requires broader public understanding. Janashantha and Abegunawardhana (2025) describe Sri Lanka as a biodiversity hotspot with over 3,200 endemic species, while noting that ecotourism often focuses on charismatic vertebrates such as elephants and leopards, leaving endemic invertebrates and microhabitats underrepresented. This creates a need for conservation communication that values all biodiversity, not only visible flagship species.

> **Example:** A visitor or community member may understand rules about elephants or leopards but may not know why freshwater crabs, land snails, amphibians, pollinators, wetland plants, or microhabitats also matter. If conservation information is technical, only in English, or presented through written signage, people may miss the full ecological meaning of protected-area guidance.

Urban wetland restoration presents another communication challenge. In Kolonnawa, residents recognized the role of wetlands in reducing flood risks, but restoration programs faced opposition because communities feared loss of land, housing, income, and social networks. This shows that ecological restoration cannot be successful unless local knowledge, livelihood concerns, and community participation are clearly addressed.

Sustainable land use also matters at the household level. The homegarden source highlights how Sri Lankan homegardens can provide food, income, pollinator support, nutrient cycling, and ecosystem services. These community-based land systems show that biodiversity-friendly practices can support both household resilience and environmental sustainability when people receive practical guidance.

Use Case Narrative: Kamal's Journey from Conflict and Confusion to Ecosystem Stewardship

To illustrate the human stakes and potential solutions, consider the story of Kamal, a 46-year-old smallholder farmer living near a protected forest and wetland area in Sri Lanka. His family depends on cultivation, small livestock, and seasonal income, but elephant movement, crop damage, changing land use, and unclear conservation notices make daily decisions stressful.

Kamal respects wildlife and understands that forests and wetlands are important. At the same time, he worries when elephants damage crops, when children walk home after evening classes, or when notices about forest boundaries and restricted activities are shared in formal language that he does not fully understand.

One season, Kamal hears that a new restoration and conservation program will begin near his village. Some neighbours say the program will reduce flooding and protect biodiversity. Others fear that it may restrict farming, reduce access to common resources, or lead to relocation. Official notices mention habitat restoration, buffer zones, biodiversity corridors, invasive species control, and compensation procedures, but the practical meaning is unclear.

Kamal searches for information, but the available material is scattered across posters, office announcements, social media posts, and technical documents. Some information is in English, while local explanations are incomplete or contradictory. Kamal speaks Sinhala and understands some Tamil phrases, but he does not feel confident asking officials about boundaries, reporting steps, or compensation eligibility.

Over the next few weeks, confusion grows. Some farmers take their own protective measures, while others ignore official guidance because they do not understand it. A few residents oppose restoration because they fear losing land or livelihoods. Community meetings are held, but older residents, women, and lower-income households do not always feel comfortable speaking or asking questions.

The situation becomes more serious after an elephant enters nearby fields and damages crops. Kamal wants to report the incident and request support, but he is unsure which office to contact, what evidence is needed, or whether reporting will create problems for his family. He also wants to know how to protect crops without harming wildlife.

A turning point comes when a local conservation group introduces a pilot program using VosynCore for ecosystem and wildlife communication. The facilitator explains that Kamal can speak into a mobile browser in Sinhala, Tamil, or English and receive clear voice guidance about wildlife safety, wetland restoration, protected-area rules, biodiversity awareness, reporting steps, and available support services.

For the first time, conservation information feels practical. VosynCore explains a wetland restoration notice in simple spoken language, clarifies why buffer zones matter, and provides a step-by-step guide for reporting crop damage. Instead of presenting conservation as a set of restrictions, it explains how healthy habitats can reduce conflict, protect flood-retention areas, and support long-term livelihoods.

The system also supports two-way questions. Kamal asks, "What should I do if elephants enter my field at night?" VosynCore provides safe, non-harmful response guidance and explains when to contact local authorities. He asks what a biodiversity corridor means, and the system explains that connected habitat can help wildlife move more safely and reduce pressure on farms.

This narrative underscores why VosynCore is relevant to SDG 15. Although Kamal's story is fictional, it reflects real communication barriers documented in Sri Lanka: human-elephant conflict, habitat

degradation, wetland loss, community concerns about restoration, and uneven biodiversity awareness.

By replacing complex conservation notices and technical restoration explanations with localized, interactive voice guidance, VosynCore helps transform ecosystem information into practical community action. It supports the human layer of land conservation: understanding, trust, participation, and coexistence.

Solution Overview: VosynCore's AI-Powered Access to Biodiversity and Land Conservation Information

VosynCore is an AI-driven platform designed to bridge language gaps in real time, enabling biodiversity guidance, wetland restoration information, wildlife safety messages, protected-area rules, ecotourism instructions, land-use notices, and ecosystem stewardship materials to be delivered in languages and formats communities understand best. It combines real-time translation, voice interaction, and localized explanations for Sinhala, Tamil, English, and local speech patterns.

This core capability directly addresses the barriers that keep communities like Kamal's from acting on ecosystem information. By transforming technical restoration plans, wildlife-conflict forms, protected-area notices, ecotourism rules, and biodiversity campaigns into voice-based, inclusive experiences, VosynCore ensures that vital land conservation information speaks the user's language and gives clear next steps.

Key Innovations of VosynCore:

- **Real-Time Voice Translation for Ecosystem Guidance:** VosynCore can take written or audio content from wildlife notices, protected-area rules, wetland restoration plans, biodiversity campaigns, or ecotourism guidance and instantly convert it into clear spoken guidance in Sinhala, Tamil, English, or local speech patterns.
    - **Example:** When a local authority issues a notice about wetland restoration or wildlife movement, VosynCore can generate a voice message explaining the affected area, timing, basic safety guidance, community meeting details, and practical reporting steps in a form that households can understand quickly.
- **Two-Way Conversational Support:** Ecosystem decisions often require clarification. Residents may need to ask what a buffer zone means, how to report elephant damage, why a wetland should not be filled, or what activities are allowed inside a protected area.
    - **Example:** If Kamal does not understand the difference between a wildlife corridor, a restricted forest boundary, and a community restoration area, he can ask in his own language and receive a simple explanation that supports informed decision-making.
- **Localized Technical Explanation:** The platform can translate technical terms such as biodiversity hotspot, endemic species, habitat fragmentation, ecosystem restoration, invasive species, wetland retention, human-wildlife conflict, and protected-area buffer zone into plain-language explanations that fit local context and community practice.

    - **Example:** Instead of saying only that a wetland provides ecosystem-based disaster risk reduction, the system can explain that wetlands hold excess rainwater, reduce flooding, support plants and animals, and protect nearby communities when managed fairly.

- **Community Conservation Awareness Support:** VosynCore can support local officers, schools, farmer groups, women's groups, ecotourism operators, conservation organizations, and youth groups by translating biodiversity awareness material, wildlife safety guidance, and restoration instructions during outreach sessions.

    - **Example:** In a village meeting, community members can receive local-language guidance on human-elephant safety, wetland protection, homegarden biodiversity, ecotourism conduct, reporting illegal activities, and participating in restoration without losing livelihood voice.

- **Data Privacy and Security:** Wildlife reports, land concerns, household economic losses, location details, and community feedback must be handled responsibly. VosynCore includes privacy safeguards, responsible data handling, and escalation pathways to human officers or certified partners when a question requires official verification or case management.

Together, these features ensure that the technology is practical, respectful, and culturally grounded. By combining real-time multilingual support with trusted community pathways, VosynCore acts as a communication bridge between conservation agencies, local authorities, community organizations, ecotourism partners, and households.

Alignment with SDG Targets & Global Frameworks

Through its context-aware AI translation platform, VosynCore directly tackles the linguistic, literacy, and participation barriers that prevent communities from understanding and acting on ecosystem protection information. The platform functions as an enabling layer for SDG 15 by improving access to biodiversity awareness, wildlife safety guidance, wetland restoration communication, and land stewardship support.

SDG 15 Targets Addressed:

Target 15.1: Conserve, restore and sustainably use terrestrial and inland freshwater ecosystems

Forests, wetlands, grasslands, homegardens, and inland freshwater ecosystems become more protected when communities can understand their ecological functions, permitted uses, restoration plans, and reporting pathways. Without accessible communication, people may see conservation as unclear restriction rather than shared stewardship.

- **VosynCore's Role:** By translating wetland protection notices, forest-use guidance, restoration instructions, and biodiversity awareness material into localized voice communication, VosynCore helps communities understand ecosystem services and take practical action.

Target 15.2 and 15.3: Promote sustainable forest management and combat land degradation

Sustainable land management depends on clear guidance around soil protection, homegarden practices, forest restoration, invasive species control, and land-use planning. Communities may support restoration in principle but hesitate when livelihood impacts or land rights are unclear.

- **VosynCore's Role:** The platform can explain restoration steps, sustainable land-use practices, community consultation materials, and land-degradation prevention guidance in plain local language, supporting trust and participation.

Target 15.5, 15.7 and 15.9: Protect biodiversity, reduce wildlife threats and integrate ecosystem values into planning

Biodiversity protection requires action on threatened species, human-wildlife conflict, illegal wildlife activities, microhabitat protection, and conservation planning. Technical terms and fragmented institutional communication can limit community participation.

- **VosynCore's Role:** By making wildlife safety guidance, biodiversity education, protected-area rules, ecotourism instructions, and reporting steps understandable, VosynCore supports more inclusive participation in ecosystem protection and biodiversity governance.

Cross-Reference to SDG 13: Climate Action and SDG 11: Sustainable Cities and Communities

Although this use case is primarily aligned with SDG 15: Life on Land, it also connects to SDG 13 because wetlands, forests, homegardens, and biodiversity-rich landscapes support climate resilience, flood regulation, and ecosystem-based adaptation. It also connects to SDG 11 where urban wetlands reduce flood risk and support more resilient cities. However, this report keeps the main focus on land ecosystems, biodiversity, habitat protection, and community stewardship.

Alignment with Impact Measurement Frameworks

Vosyn is deeply committed to tracking, auditing, and reporting its Social Return on Investment (SROI) in compliance with institutional investor-grade data standards. Our data architecture is built to map cleanly against international Environmental, Social, and Governance (ESG) standards, including IRIS+ managed by the Global Impact Investing Network, the Global Reporting Initiative (GRI), and Sustainability Accounting Standards Board (SASB) infrastructures. By anchoring forward-looking models to standardized global frameworks, Vosyn ensures operational transparency, data comparability, and credibility for impact investors and institutional partners.

For more information on how we meet compliance frameworks, visit our ***Compliance Document.***

KPI Mapping & Other Frameworks

We have identified specific KPIs to measure VosynCore's performance and impact. The table below shows example KPIs, their linkage to relevant SDG 15 targets and indicators, and alignment with standard metrics from IRIS+ and other frameworks such as GRI and SASB where applicable:

<	SDG 15 Target & Indicator	Aligned IRIS+ Metric	Other Framework Alignment
?	**SDG 15.1** Access to affordable, reliable and modern energy services. **SDG 15.b** Expanded sustainable energy infrastructure.	**Client Individuals:** Total number of households or enterprises served by the program.	**GRI 203-2 (Significant indirect economic impacts):** Reports beneficiaries of services and social benefit; **HIPSO People Benefited:** Aligned with development-finance comparison.
x	**SDG 15.1** Reliable and modern energy services. **SDG 15.3 Improved energy efficiency.**	**Outcome Indicator (custom):** Measures change in knowledge, comprehension, and ability to act on service information.	**GRI 417 (Product/service information):** Ensures information is understandable; **GRI 413-1:** Measures local program effectiveness.
m	**SDG 15.2** Increase the share of renewable energy. **SDG 13.2** Integrate climate action into planning and programs.	**Client Outcome Metric:** Tracks readiness, confidence, and informed decision-making among served households.	**GRI 302 (Energy):** Reflects energy-efficiency and renewable energy engagement; **SASB:** Energy management and consumer access.

I	**SDG 15.3** **Improve energy efficiency.** **SDG 12.2** **Efficient use of resources.**	**Resource Efficiency Outcome:** Tracks behaviour change connected to reduced energy waste and household cost management.	**GRI 302-4:** Supports reporting on energy efficiency; aligns with climate and consumer education.
	SDG 15.a International cooperation and clean energy access. **SDG 10.2** Reduced exclusion through accessible services.	**Product Accessibility Metric:** Demonstrates inclusivity and reach to underserved language groups.	**GRI 417:** Ensures critical information is provided in accessible formats; **SASB:** Technology-accessibility indicators.
v	**SDG 15.b** Expanded sustainable energy services. **SDG 13.3** Improved climate education and awareness.	**Stakeholders Trained:** Adapted from IRIS+ training metrics to capture local capacity-building.	**GRI 404-1:** **Reports training provided;** **GRI 413-1:** Shows strengthened community capacity and participatory delivery.

Table 1: VosynCore SDG 15 Performance Matrix, including KPI mapping, UN indicator harmonization, and institutional ESG framework alignment for the Sri Lanka Biodiversity and Land Conservation Communication Pilot.

By establishing this standardized, transparent KPI framework, VosynCore bridges the operational gap between ecosystem protection needs and institutional impact reporting. This data-driven mapping ensures that every translated restoration notice, clarified wildlife instruction, and completed conservation inquiry is measurable, auditable, and benchmarked against recognized international standards.

For public agencies, conservation partners, ecotourism operators, community organizations, and impact investors, this matrix provides a derisked pathway for converting language access into

measurable improvements in biodiversity awareness, wildlife-conflict reporting, wetland restoration participation, and community stewardship.

For more information on the compliance frameworks that govern our operations, visit our ***Compliance Document.***

Impact Pathway

VosynCore's Theory of Change (TOC) for SDG use cases outlines how inputs and activities translate into outcomes that contribute to long-term change. For SDG 15, the pathway begins with localized language access and leads toward improved ecosystem-service comprehension, safer human-wildlife coexistence, wetland restoration participation, biodiversity-friendly behaviour, and more inclusive land conservation governance. Key assumptions, risks, and mitigation strategies will be documented through pilot monitoring and learning.

Learning Agenda, Pilot Roadmap & Scalability Strategy

Sri Lanka Biodiversity and Land Conservation Communication Pilot

VosynCore's SDG 15 initiative is not a one-off deployment; it is conceived as a learning journey that will inform scaling and broader application. The pilot will test how voice-based translation and conversational guidance can help households, farmers, informal-settlement residents, local officers, conservation groups, and ecotourism partners understand ecosystem information more accurately and act on it more confidently.

Learning Agenda

We have formulated key learning questions to guide our SDG 15 pilot and inform future scaling. These questions ensure that the technology supports practical ecosystem stewardship, wildlife safety, wetland restoration communication, and trusted community engagement without overstating the role of AI in solving physical land-use conflicts or habitat loss.

1. Effectiveness in Ecosystem Information Comprehension and Community Action: How much does localized voice guidance improve community understanding of wetland protection, protected-area rules, wildlife safety, restoration plans, and biodiversity information compared to standard written or broadcast messages?

- **Goal:** Compare comprehension scores, correct interpretation of conservation notices, wildlife-reporting understanding, and follow-up actions among communities using VosynCore and communities receiving standard ecosystem communication.
- **Example:** Is there a measurable increase in households correctly explaining what a wetland restoration notice means, how to report elephant-related damage, and which biodiversity protection actions are safe and useful after receiving localized voice guidance?

2. Human-Wildlife Coexistence and Trust in Conservation Guidance: How does conversational support affect household confidence, perceived fairness, and willingness to follow non-harmful wildlife safety steps or participate in habitat restoration?

- **Goal:** Track changes in trust, perceived usefulness, confidence in reporting, and ability to identify official support pathways for wildlife conflict, protected-area rules, and community restoration activities.

- **Example:** Do households that receive voice-based explanation and question-answer support show higher willingness to report human-elephant conflict through official channels or attend conservation information sessions?

3. Community Integration and Institutional Workflow: How effectively do local officers, conservation partners, ecotourism operators, schools, and community groups incorporate VosynCore into biodiversity and land conservation communication workflows?

- **Goal:** Observe partner workflows to determine whether the tool reduces explanation time, improves clarity, increases completed reports, and strengthens participation in restoration, ecotourism, or biodiversity awareness programs.

- **Example:** Does VosynCore reduce the need for repeated manual explanation during village meetings or protected-area outreach by allowing facilitators to provide accurate, localized guidance in real time?

To answer these questions, the Monitoring, Evaluation, and Learning (MEL) team will employ a mixed-methods approach. Quantitatively, we will use baseline surveys, comprehension quizzes, app usage analytics, reporting-data tracking, and community action checklists. Qualitatively, we will conduct focus groups with farmers, informal-settlement residents, women's groups, youth groups, ecotourism workers, conservation officers, and local leaders to understand trust, usability, and cultural appropriateness.

For more information, please visit our ***"Impact Pathway" document.***

Pilot Roadmap

The pilot will proceed in iterative stages to ensure that the technology is safely and effectively integrated into biodiversity awareness, wetland restoration communication, human-wildlife coexistence support, protected-area guidance, and community land stewardship workflows.

Months 0-3 (Preparation and Co-Design): Establish Partnerships and Foundations

- Finalize formal agreements with conservation agencies, local authorities, wildlife officers, community organizations, schools, farmer groups, women's groups, ecotourism partners, and wetland restoration stakeholders in selected pilot communities.

- Conduct co-design workshops with farmers, communities living near protected areas, informal-settlement residents, ecotourism workers, youth groups, and local facilitators to capture terminology for wetlands, forests, endemic species, wildlife conflict, habitat restoration, buffer zones, reporting, and compensation pathways.

- Train facilitators in inclusive communication, responsible AI use, privacy safeguards, referral boundaries, and the limits of VosynCore in legal land decisions, wildlife case management, compensation approval, and technical ecological assessments.

- **Milestones:** Secure partner approvals; configure the initial VosynCore version with Sinhala, Tamil, English, and local speech-pattern support; complete baseline surveys on biodiversity awareness, wetland knowledge, wildlife-conflict reporting, restoration trust, and conservation communication needs.

Months 4-9 (Implementation and Monitoring): Field Deployment and Feedback Loops

- Roll out mobile or browser access through community organizations, local officers, schools, farmer groups, ecotourism networks, conservation sessions, and wetland restoration meetings.

- Implement real-time monitoring of translation accuracy, user comprehension, wildlife-safety instruction clarity, wetland guidance interpretation, protected-area rule explanation, and completed conservation or conflict reports.

- Facilitate monthly learning circles with households and partner organizations to adapt voice prompts, technical explanations, and escalation pathways based on actual field usage.

- **Milestones:** Achieve target engagement within pilot communities; collect interim data on comprehension, trust, wildlife reporting, restoration participation, biodiversity awareness, and facilitator workload.

Months 10-12 (Evaluation and Scaling Prep): Impact Assessment and Synthesis

- Execute endline surveys and final focus groups to measure changes in ecosystem knowledge, human-wildlife coexistence confidence, wetland restoration participation, biodiversity-friendly behaviour, reporting confidence, and participation in conservation decision-making.

- Synthesize findings into a Biodiversity and Land Conservation Communication report for public agencies, conservation partners, local authorities, ecotourism organizations, community groups, and potential scaling partners.

- Refine AI models and guidance content based on the pilot's linguistic, cultural, ecological, and workflow data to prepare for expansion into additional districts and ecosystem-related use cases.

- **Milestones:** Publish a validated impact report aligned with SDG 15 KPIs; finalize the roadmap for Phase 2 expansion into human-wildlife coexistence support, wetland restoration communication, protected-area guidance, biodiversity education, and sustainable land-use communication.

Scalability Strategy

Looking beyond the pilot, our ambition is to scale both vertically, by deepening integration with conservation, disaster-risk, agriculture, ecotourism, and local governance institutions, and horizontally, by adapting the model to new districts, languages, habitats, and community land-use settings.

- **Institutional Integration:** We aim to embed VosynCore into existing biodiversity education, wildlife-conflict reporting support, wetland restoration communication, protected-area outreach, ecotourism guidance, and community land stewardship programs.
    - **Example:** Following a successful pilot, Vosyn will collaborate with conservation agencies, local authorities, schools, farmer organizations, and ecotourism partners to incorporate localized voice guidance into wildlife notices, wetland restoration meetings, and biodiversity awareness campaigns.
- **Linguistic and Geographic Replicability:** Our modular architecture allows for the rapid addition of regional speech patterns and sector-specific terminology without rebuilding the core infrastructure.
    - **Example:** Once the model is stabilized for Sinhala and Tamil ecosystem content in one pilot district, the same framework can be adapted for Dry Zone human-elephant conflict areas, Western Province wetlands, Central Highland biodiversity areas, and rural homegarden-based communities.
- **API-First Partnership Model:** To achieve broader reach, we will offer VosynCore as an API for conservation portals, protected-area information systems, wildlife reporting tools, ecotourism platforms, local government pages, and environmental education systems.
- **Example:** A conservation agency or local authority platform can integrate VosynCore's translation API so that restoration notices, protected-area rules, biodiversity guidance, and wildlife safety instructions are automatically available as localized voice guidance.
- **Community-Led Expansion:** We leverage a train-the-trainer model where successful pilot facilitators, farmers, women's groups, youth leaders, ecotourism workers, and local officers become champions for new deployment sites.
    - **Example:** Community facilitators who learn to use VosynCore can support onboarding sessions in neighbouring villages, wetland settlements, or tourism-adjacent communities, ensuring that scaling is driven by trusted local actors rather than external consultants alone.

This strategy ensures that VosynCore evolves from a localized intervention into a broader standard for linguistic inclusion in biodiversity and land conservation communication, ensuring that language and literacy are never barriers to ecosystem protection and sustainable coexistence.

Modelled Projections

As VosynCore moves toward field deployment, these evidence-based projections model potential impact on ecosystem comprehension, biodiversity awareness, wildlife-conflict reporting, wetland restoration participation, and institutional communication efficiency.

Projected Outcomes (Short to Medium Term):

In the initial 12-24 months of deployment, the pilot will establish a baseline of community understanding, conservation readiness, and local engagement. Expected outcomes include:

- **Increased Ecosystem Information Comprehension:** A projected 40% increase in households correctly understanding wetland restoration notices, protected-area rules, wildlife safety guidance, biodiversity messages, and sustainable land-use instructions.
- **Improved Conservation Participation Readiness:** A projected 25% increase in households completing a wildlife-conflict inquiry, attending a biodiversity information session, joining a restoration meeting, or accurately identifying next steps for conservation participation.
- **Stronger Biodiversity-Friendly Practice:** A measurable increase in safe wildlife reporting, reduced wetland dumping or filling, careful ecotourism conduct, homegarden biodiversity support, and better understanding of protected-area responsibilities.
- **Operational Efficiency:** Local officers, conservation teams, and community facilitators reduce repeated clarification requests and deliver more consistent ecosystem guidance.

Projected Outcomes (Medium to Long Term):

Looking toward 2030, VosynCore can support more inclusive ecosystem communication systems:

- **Sustained Ecosystem Literacy:** Households maintain clearer understanding of biodiversity, habitat restoration, wetland functions, safe wildlife practices, land degradation risks, and conservation support pathways.
- **Inclusive Biodiversity Participation:** Farmers, low-income urban residents, women, youth, ecotourism workers, and underserved communities are better able to participate in conservation and restoration planning.
- **Climate- and City-Aligned Ecosystem Communication:** Communities receive accessible information about wetlands, forests, homegardens, flood regulation, habitat protection, and ecosystem-based adaptation.
- **Global Scaling of Inclusive Ecosystem Infrastructure:** VosynCore serves as a replicable model for language-accessible biodiversity and land conservation communication across multilingual communities.

Conclusion: High-Potential Innovation for SDG 15

VosynCore addresses one of the most overlooked barriers to life on land: the communication gap between technical ecosystem systems and everyday community decision-making. In Sri Lanka, human-elephant conflict, wetland degradation, biodiversity conservation gaps, and sustainable land-use needs show that ecosystem protection depends on both ecological management and understanding.

This initiative does not restore forests by itself, resolve land ownership disputes, or replace conservation officers. Instead, it strengthens the reach of public agencies, community

organizations, ecotourism partners, and conservation groups by ensuring that essential information is available in languages and formats communities can act on.

Key Takeaways:

- **Systemic Force Multiplier:** VosynCore makes wetland notices, wildlife guidance, biodiversity education, protected-area rules, and restoration instructions understandable to communities.
- **Measurable Social Return on Investment (SROI):** The platform converts language accessibility into measurable outcomes such as improved comprehension, safer reporting, stronger restoration participation, and better community stewardship.
- **Targeted Equity and Ecosystem Impact:** Localized audio guidance supports farmers, informal-settlement residents, women, low-literacy users, ecotourism workers, and underserved communities.
- **Rigorous Institutional Alignment:** The deployment roadmap aligns with SDG Targets 15.1, 15.2, 15.3, 15.5, 15.7, and 15.9, while supporting IRIS+, GRI, and SASB-aligned reporting.

Achieving SDG 15 requires systems that help people understand ecosystem values, trust official guidance, report wildlife issues safely, and participate in planning. VosynCore directly addresses the hidden barrier of language and literacy in conservation communication.

Our SDG 15 initiative offers a high-potential innovation that can help ecosystem protection speak every person's language, empowering communities to conserve biodiversity, restore land, reduce conflict, and participate in a more sustainable future.

References

Institutional References (UN & Global Reports)

United Nations Department of Economic and Social Affairs. (2025). *Goal 15: Life on Land*. United Nations Sustainable Development Goals. https://sdgs.un.org/goals/goal15

Convention on Biological Diversity. (2024). *Global Biodiversity Framework resources*. https://www.cbd.int/

International Union for Conservation of Nature. (2024). *Ecosystem restoration and species conservation resources*. https://www.iucn.org/

Scientific & Academic References

Wickramarachchi, P. I., Siriwardena, B. P., Vidanapathirana, N. P., & Bandusekara, B. S. (2026). Economic determinants of human-elephant conflict: Evidence from Lunugamwehera, Sri Lanka. *Journal of Agriculture and Value Addition*, *9*(1), 63-89.

Janashantha, A. S., & Abegunawardhana, P. D. (2025). The role of ecotourism in protecting Sri Lanka's endemic species and biodiversity hotspots: A comprehensive review. *IRC-OUSL 2025*.

Lansakara, D. C. J., Le De, L., Petterson, M., & Wickramasinghe, D. (2026). Ecosystem-based DRR in urban Sri Lanka: The socio-political dynamics of wetland restoration. *Environmental Hazards*.

Peiris, P. U. S., & Wijesekara, R. G. S. (2025). Role of homegardens in ensuring food and nutritional security and uplifting bioeconomy of Sri Lankan communities. *YSF Thematic Publication 2025*, NASTEC.